Matías Javier Vega Paredes

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Mechanical Engineering Student | CFD & Numerical Modelling

Profile

Final-semester mechanical engineering student with experience in ANSYS Fluent studies, numerical modelling and experimental flow analysis in Python and MATLAB. Seeking graduate CFD and aerodynamic development roles; professional degree expected Jan 2027.

Education

Universidad de Chile **Santiago, Chile**
Professional Degree in Mechanical Engineering *Expected Jan 2027*
Ingeniero Civil Mecánico. 5.5-year programme (11 semesters): four-year *Licenciatura en Ciencias de la Ingeniería, mención Mecánica*, followed by 1.5 years of professional specialisation including a research thesis. Current average: 5.6/7.0 (Chilean scale).

Technical Skills

CFD: ANSYS Fluent; geometry, inflation-layer meshing, skewness/orthogonal quality, model setup and post-processing
Numerical analysis: Python (NumPy, FFT), MATLAB; RK4, frequency-domain analysis, PIV processing
CAD: SolidWorks, Fusion 360; structural analysis in SolidWorks Simulation
Other: Arduino/C, QuickDAQ, Power Apps, Power Automate; Excel (intermediate)

Selected Simulation & Research Projects

F1-Style Rear-Wing CFD Study **2025**
Computational Fluid Mechanics course project *Universidad de Chile*

- Designed a two-element rear wing and compared open/closed drag reduction system (DRS) configurations in ANSYS Fluent; generated inflation-layer meshes, assessed skewness/orthogonal quality, selected models and analysed results.
- Compared the simulated drag-coefficient reduction on opening DRS (65%) with the 63% reported in reference literature, alongside the lift/drag trade-off.

Experimental Fluid Mechanics Thesis – PIV **Mar 2026–Present**
LEAF-NL, Universidad de Chile *Santiago, Chile*

- Developed a multi-pass PIV workflow in Python (NumPy FFT) and MATLAB for 240 fps recordings: image calibration and preprocessing, cross-correlation, vector validation and conversion to physical units.
- Calculated velocity/vorticity fields, circulation and their temporal evolution to study undular-bore interaction with partially submerged objects.
- Designed and built a wave-generation system with a DC motor, 3D-printed parts and timing belts; integrated encoder/current measurements and Arduino control through a computer interface.

F1 Porpoising Model **2025**
Numerical Methods course project *Universidad de Chile*

- Implemented a two-degree-of-freedom heave/pitch model in Python and MATLAB, coupling suspension dynamics to ride-height-dependent aerodynamic loading and a simplified stall function; solved using RK4 and analysed the response via FFT.

Engineering Experience

HORSE Chile SpA **Jan–Feb 2026**
Maintenance Engineering Intern *Automotive powertrain manufacturing*

- Evaluated cooling conditions at a 3 kW laser-welding station against manufacturer flow/temperature requirements; recommended increased overnight chiller operation, after which hot-day faults were observed around 14:00 rather than 10:00.
- Used accelerometer measurements and 40/80 Hz spectra to diagnose motor misalignment on a transmission test bench; inspection confirmed displaced shims/loose fasteners, and corrective work reduced observed vibration.
- Analysed ten years of demand data for approximately 860 critical spare parts to recalculate reorder points and safety stocks; automated replenishment notifications and follow-up alerts in Power Automate.

Boetek **Jan–Feb 2025**
Engineering Intern *Thermal engineering*

- Developed a Python workflow to notify on-call technicians by automated call/text when the boiler predictive-maintenance system reported a failure.
- Measured pump flow, pipe temperatures and combustion gases; compared operating conditions and identified 50% load as the best-efficiency point for the tested boiler.

Additional Engineering Projects

Furnace Waste-Heat Recovery Feasibility Study **2025**
Engineering in Thermofluids project with HORSE Chile *Universidad de Chile*

- Contributed to basic-engineering feasibility calculations for preheating process oil with furnace exhaust heat; focused on exhaust-gas flow, pump selection, pressure losses and economic assessment.

U Racing Team, Formula SAE **Late 2024–Present**
Suspension & Brakes Lead (since 2026) *Universidad de Chile*

- Coordinate four team members and liaise with the other vehicle subsystem teams as Suspension & Brakes Lead.
- Designed steering geometry and suspension layouts in SolidWorks for a Formula SAE vehicle under development.
- Selected upright material and supported CNC manufacturing planning with the faculty workshop technician, including tooling selection and preparation.

Languages & Mobility

Spanish: Native. English: University coursework completed (FCFM, Levels 1–5).
Available to relocate to the UK or Europe.